NATIONAL UNIVERSITY OF SINGAPORE

MA1513 – EXAM (90 minutes)

AY 2024/2025 Semester II

INSTRUCTIONS TO STUDENTS

(a) There are 15 MCQ and 4 written questions in this exam with a total of 90 marks.

(b) The MCQ should be answered and submitted through Examplify.

(c) For the written questions (Q16 to Q19), use the exam answer booklet to write the answers.

- Read the instructions on the cover page of the answer booklet carefully.
- Indicate your student number clearly on the cover page of the answer booklet.
- Your answer booklet will be collected at the end of the exam.
- You do not need to scan and submit the PDF version to Canvas.

(d) Students are not allowed to exit Examplify before the end of the exam. At the end of the 90 minutes, Examplify will close automatically, and you should immediately stop writing.

(e) This is an open book examination.

(f) You may use MATLAB and any calculator. However, you should explain clearly how the answers of the written questions are obtained.

MA1513 Exam (Examplify)

AY2024/2025 Semester II

MCQ/FIB (15 questions; 3 marks each)

1. Let **Ax** = **b** be a non-homogeneous linear system where **R** is the reduced row echelon form (inclusive of the augmented column) of the augmented matrix (**A | b**). Which of the following statements must be true? (Pick all correct options)
   i. If every column of **R** is a pivot column, then **Ax** = **b** is inconsistent.
   ii. If **R** has a non-pivot column, then **Ax** = **b** has infinitely many solutions.
   iii. If **R** has a zero row, then **Ax** = **b** is consistent.
   iv. If every row of **R** has a leading entry, then **Ax** = **b** has exactly one solution.
   v. If **R** does not have a leading entry in the last column, then **Ax** = **b** is consistent.

2. Which of the following are equivalent conditions for an n x n matrix **A** to be non-singular? (Pick all correct options)
   i. The n rows of **A** are linearly independent.
   ii. Any row echelon form of **A** has n pivot columns.
   iii. The column space of **A** is $\mathbf{R}^n$.
   iv. The determinant of **A** is 1.
   v. The non-homogeneous system **Ax** = **b** has only the trivial solution.

3. Given matrix $\mathbf{A} = \begin{pmatrix} a & b & c \\ i & j & k \\ x & y & z \end{pmatrix}$ has a non-zero determinant n. Which of the following are correct? (Pick all correct options)
   i. $\begin{vmatrix} a & b & c \\ 2i & 2j & 2k \\ x & y & z \end{vmatrix} = 2^3 n$
   ii. $\begin{vmatrix} x & y & z \\ a & b & c \\ i & j & k \end{vmatrix} = -n$
   iii. $\begin{vmatrix} z & y & x \\ k & j & i \\ c & b & a \end{vmatrix} = n$
   iv. $\begin{vmatrix} a & 3i & 5x \\ b & 3j & 5y \\ c & 3k & 5z \end{vmatrix} = 15n$
   v. $\begin{vmatrix} a-i & b-j & c-k \\ i-x & j-y & k-z \\ x-a & y-b & z-c \end{vmatrix} = n$

4. A is an m x n matrix which is full rank. Which of the following are correct? (Pick all correct options.)
   i. If m < n, it is always possible to add an additional row to **A** so that the resulting matrix has a larger rank.
   ii. If m < n, it is always possible to add an additional column to **A** so that the resulting matrix has a larger rank.
   iii. If m > n, it is always possible to add an additional row to **A** so that the resulting matrix has a larger rank
   iv. If m > n, it is always possible to add an additional column to **A** so that the resulting matrix has a larger rank.
   v. If m = n, adding an additional row or column to **A** will result in a matrix with a larger rank.

5. Which of the following represent the plane x – y + 2z = 0 in the xyz-space? (Pick all correct options)
   i. span{ (1, - 1, 2), (0, 0, 0) }
   ii. span{ (1, 1, 0), (0, 2, 1), (2, 0, -1) }
   iii. span{ (1, 3, 1), (3, 9, 3) }
   iv. span{ (3, 1, -1), (3, -1, -2) }
   v. span{ (1, 0, 0), (0, -1, 0), (0, 0, 2)}

6. Which of the following sets are linearly independent? (Pick all correct options)
   i. { (10, 11, 12), (100, 121, 144) }
   ii. { (1, 5, 1, 3), (1, 3, 1, 5), (5, 1, 3, 1), (3, 1, 5, 1) }
   iii. { (3, 6, 9), (13, 16, 19), (23, 26, 29) }
   iv. { (0, 1, 2, 3), (0, 0, 1, 2), (0, 0, 0, 3) }
   v. { (1, 0, 1, 0), (0, 2, 0, 2), (3, 3, 3, 3) }

7. A 3 x 5 matrix **C** has columns $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$, $\mathbf{c}_4$, $\mathbf{c}_5$ arranged from left to right. Suppose the reduced row echelon form of **C** is $\begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$. Which of the following sets are bases for the column space of **C**? (Pick all correct options)
   i. { $\mathbf{c}_1$, $\mathbf{c}_3$, $\mathbf{c}_5$ }
   ii. { $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_5$ }
   iii. { $\mathbf{c}_1$, $\mathbf{c}_3$, $\mathbf{c}_4$ }
   iv. { $\mathbf{c}_3$, $\mathbf{c}_4$, $\mathbf{c}_5$ }
   v. Any set of 3 linearly independent vectors in $\mathbf{R}^3$.

8. Given S = { $\mathbf{u}_1$, $\mathbf{u}_2$ } is a basis for $\mathbf{R}^2$. Suppose the coordinate vector of $\mathbf{v} = \binom{-3}{6}$ with respect to S is given by $(\mathbf{v})_S = \binom{2}{-1}$ and the coordinate vector of $\mathbf{w} = \binom{6}{3}$ with respect to S is given by $(\mathbf{w})_S = \binom{-1}{2}$. Then $\mathbf{u}_1 = \binom{a}{b}$ and $\mathbf{u}_2 = \binom{c}{d}$ where a = _0_, b = _5_, c = _3_, d = _4_ (FIB)

9. Let T: $\mathbf{R}^2 \rightarrow \mathbf{R}^2$ be a linear transformation such that all the vectors along the line y = 2x is scaled by a factor 2, and all the vectors along the line y = -2x is scaled by a factor -2. Then the standard matrix of T is given by $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where a = _0_, b = _1_, c = _4_, d = _0_ (FIB)

10. Let **A** be a 4 x 4 matrix. Suppose two of its eigenvalues are 3 and 0 with corresponding eigenspaces E_3 = span$\left\{\begin{pmatrix}1\\0\\0\\1\end{pmatrix},\begin{pmatrix}0\\1\\1\\0\end{pmatrix},\begin{pmatrix}1\\2\\2\\1\end{pmatrix}\right\}$ and E_0 = span$\left\{\begin{pmatrix}0\\0\\0\\1\end{pmatrix}\right\}$.
Which of the following statements must be true? (Pick all correct options)
    i. **A** has exactly 2 eigenvalues.
    ii. The multiplicity of the eigenvalue 0 is equal to 1.
    iii. **A** can be a non-diagonalizable matrix.
    iv. Dimension of E_3 is equal to 3.
    v. $\lambda = 3$ is a repeated root of the characteristic polynomial of **A**.

11. Suppose $\mathbf{P}^{-1}\mathbf{AP} = \mathbf{D}$ where $\mathbf{P} = \begin{pmatrix}1 & 0 & 2\\1 & 0 & 1\\0 & 1 & 0\end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix}2 & 0 & 0\\0 & 2 & 0\\0 & 0 & 4\end{pmatrix}$. Which of the following are true? (Pick all correct options)
    i. $\begin{pmatrix}2 & 1 & 0\\1 & 1 & 0\\0 & 0 & 1\end{pmatrix}^{-1} A \begin{pmatrix}2 & 1 & 0\\1 & 1 & 0\\0 & 0 & 1\end{pmatrix} = \begin{pmatrix}4 & 0 & 0\\0 & 2 & 0\\0 & 0 & 2\end{pmatrix}$
    ii. $\begin{pmatrix}1 & 0 & 4\\1 & 0 & 2\\1 & 2 & 0\end{pmatrix}^{-1} A \begin{pmatrix}1 & 0 & 4\\1 & 0 & 2\\1 & 2 & 0\end{pmatrix} = \begin{pmatrix}2 & 0 & 0\\0 & 2 & 0\\0 & 0 & 8\end{pmatrix}$
    iii. $\begin{pmatrix}1 & 0 & 2\\1 & 0 & 1\\0 & 1 & 1\end{pmatrix}^{-1} A \begin{pmatrix}1 & 0 & 2\\1 & 0 & 1\\0 & 1 & 1\end{pmatrix} = \begin{pmatrix}2 & 0 & 0\\0 & 2 & 0\\0 & 0 & 4\end{pmatrix}$
    iv. $\begin{pmatrix}1 & 2 & 4\\1 & 1 & 2\\1 & 0 & 0\end{pmatrix}^{-1} A \begin{pmatrix}1 & 2 & 4\\1 & 1 & 2\\1 & 0 & 0\end{pmatrix} = \begin{pmatrix}2 & 0 & 0\\0 & 4 & 0\\0 & 0 & 4\end{pmatrix}$
    v. $\begin{pmatrix}2 & 2 & 1\\2 & 1 & 1\\1 & 0 & 2\end{pmatrix}^{-1} A \begin{pmatrix}2 & 2 & 1\\2 & 1 & 1\\1 & 0 & 2\end{pmatrix} = \begin{pmatrix}2 & 0 & 0\\0 & 4 & 0\\0 & 0 & 2\end{pmatrix}$

12. Given a subspace V = span{ (1, 1, 0, 0), (0, 0, 1, 1) } and a vector **u** = (3, -1, 2, -4) in $\mathbf{R}^4$. The vector **v** in V that is closest to **u** is given by v = (a, b, c, d) where a = _1_, b = _1_, c = _-1_, d = _-1_ (FIB)

13. Given a 2 x 2 matrix **A** with only 1 eigenvalue $\lambda = -1$ and only 1 linearly independent eigenvector $\mathbf{v} = \begin{pmatrix}1\\-1\end{pmatrix}$. Suppose the linear system $(\mathbf{A} + \mathbf{I})\mathbf{u} = \begin{pmatrix}1\\-1\end{pmatrix}$ has a general solution $\mathbf{u} = \begin{pmatrix}1+s\\s\end{pmatrix}$. Which of the following are solutions of the SDE $\mathbf{y}' = \mathbf{Ay}$?
    i. $\begin{pmatrix}y_1\\y_2\end{pmatrix} = e^{-t}\begin{pmatrix}-1\\1\end{pmatrix}$
    ii. $\begin{pmatrix}y_1\\y_2\end{pmatrix} = e^{-t}\begin{pmatrix}1\\0\end{pmatrix}$
    iii. $\begin{pmatrix}y_1\\y_2\end{pmatrix} = e^{-t}\begin{pmatrix}t\\-t-1\end{pmatrix}$
    iv. $\begin{pmatrix}y_1\\y_2\end{pmatrix} = e^{-t}\begin{pmatrix}2t+1\\-2t\end{pmatrix}$
    v. $\begin{pmatrix}y_1\\y_2\end{pmatrix} = e^{-t}\begin{pmatrix}t\\-t+1\end{pmatrix}$

14. Given an SDE $\mathbf{y}' = \begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix}\mathbf{y}$. Which of the following are solutions of the SDE?

i. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \frac{1}{2}\cos t + \frac{1}{2}\sin t \\ \cos t \end{pmatrix}$

ii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \frac{1}{2}\cos t - \frac{1}{2}\sin t \\ \cos t \end{pmatrix}$

iii. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} -\sin t + \cos t \\ -2\sin t \end{pmatrix}$

iv. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \sin t + \cos t \\ 2\sin t \end{pmatrix}$

v. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^t \begin{pmatrix} \frac{1}{2}\sin t - \frac{1}{2}\cos t \\ -\sin t \end{pmatrix}$

15. Suppose **A** is a 2 x 2 non-zero matrix which is singular and diagonalizable. Which of the following are correct about the SDE **y**′ = **Ay** ? (Pick all correct options)

i. The equilibrium point can be a centre.
ii. The equilibrium point cannot be a degenerate node.
iii. The equilibrium point can be a star node.
iv. All the trajectories in the phase portrait are straight lines.
v. None of the trajectories in the phase portrait are straight lines.

Student number (9 digits starting with A) **Assigned Seat number** (3 digits)

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NATIONAL UNIVERSITY OF SINGAPORE

MA1513 Linear Algebra with Differential Equations
FINAL EXAM

Semester 2 AY2024/25

Time Allowed: 1.5 Hours

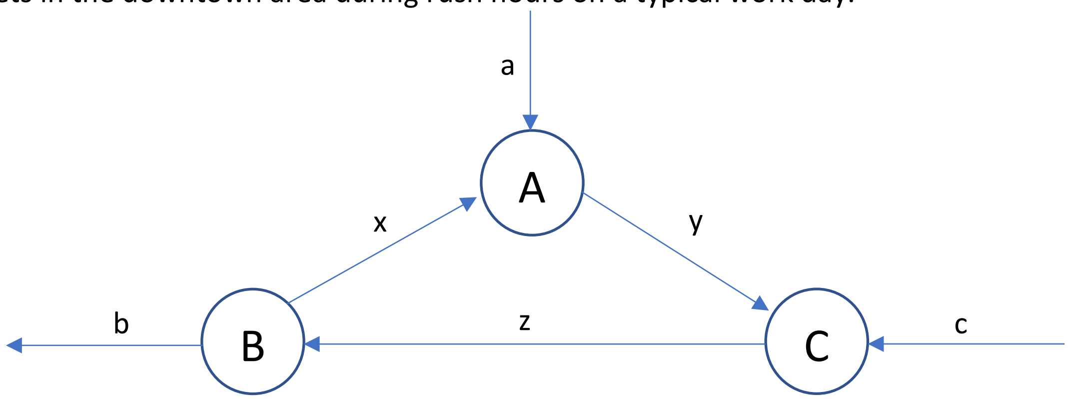
INSTRUCTIONS TO STUDENTS

1. Write down your student number and assigned seat number at the top of this page. DO NOT WRITE YOUR NAME!
2. This paper contains **FOUR** written questions and comprises **FIFTEEN** printed pages.
3. Students are required to answer the MCQ questions (Q1 to Q15) directly in Examplify, and write the answers with workings of the written questions (Q16 to Q19) in this booklet.
4. This is an OPEN BOOK exam.
5. Students are allowed to use MATLAB that are installed in their PC.
6. Students are allowed to use any calculators.

Instructions: You should write your working and answers in the spaces provided in this exam booklet only. Write the final answers in the specific boxes provided. You may use MATLAB for all the questions. But you should show how your answers are derived.

Question 16 (10 marks)

The network in the diagram shows the traffic flow (in vehicles per hour) over several one-way streets in the downtown area during rush hours on a typical work day.



(i) (3 marks) Suppose the traffic volumes for a, b, c are 100, 150 and 50 respectively. Set up a linear system **A**x = **b** with 3 equations (in the order of the junctions A, B and C) with three variables x, y, z arranged in that order. Write down the matrix **A** and vector **b** explicitly. <u>All entries of vector **b** should be positive</u>.

<u>Working space</u>

Write the matrices **A** and **b** in the box

A = **b =**

(ii) (2 marks) Find the general solution of the linear system in (i). (You may use MATLAB for the working.)

Working space

Write the general solution in the box

x =

y =

z =

(iii) (2 marks) Given the traffic volumes in (i), what is the smallest possible value of y?

Working space

Write the answer in the box

smallest y =

(iv) (3 marks) If we are not given the specific traffic volumes for a, b and c in (i), what is the condition (in terms of an equation in a, b and c) so that there are solutions for x, y, z ?

Working space

Write the equation in the box

Additional working space for Q16 (Indicate the parts clearly)

Question 17 (10 marks)

A damped mass-spring system is given by a 2^{nd} order ODE: $my'' + cy' + ky = 0$, where k is the spring constant (in Newton per meter N/m) and c is the damping constant, while m is the mass (in kg) of the ball. Three sets of readings for displacement y in meter (m), velocity y' in m/s and acceleration y" in m/s^2 are measured and recorded as follow:

Measurements	y (m) *	y' (m/s)	y" (m/s^2)
1^{st} set	0 (equilibrium position)	1.5 (downward)	-12.5
2^{nd} set	0.5 (below e. p.)	0.5 (downward)	-15
3^{rd} set	0.2 (below e. p.)	-1 (upward)	10

* Value for y is positive below the e. p., and negative above the e. p.

(i) (4 marks) Suppose the mass of the ball is 1kg. Use the above measurements to set up a linear system **Ax = b** to solve for c and k.

Working space

Write the matrices **A** and **b** in the box

A = **b =**

(ii) (2 marks) Write down the ODE that best fit the above readings.

Working space

Write the best fit ODE in the box
(give your c and k up to 2 decimal places)

(iii) (2 marks) Is the system over-damping or under-damping? Why?

Working space

Write the answer and reason in the box

Answer:

Reason (one sentence without working):

(iv) (2 marks) Convert the ODE into an SDE without solving it. (You are not required to show the working.)

Write the SDE in the box (give your answer up to 2 decimal places)

Additional working space for Q17 (Indicate the parts clearly.)

Question 18 (12 marks)

Three coffee brands A, B and C are available to the 30,000 households in city X. The monthly market shares (in terms of number of households) of the three brands is modelled by $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$ with $\mathbf{x}_n = \begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix}$ where a_n, b_n and c_n are the number of households purchasing brands A, B and C respectively in the n^{th} month. (Assuming each household only purchase one coffee brand per month.)

According to the market research, the market shares in January to April 2024 are as follow:

Brand	Jan	Feb	Mar	Apr
A	10000	11200	11740	11968
B	8000	9800	10760	11282
C	12000	9000	7500	6750

(i) (4 marks) Find the stage matrix **A**.

Working space

Write the matrix **A** in the box

A =

(ii) (3 marks) Use MATLAB command to find a matrix **P** that diagonalizes **A,** together with the diagonal matrix **D**. (You are not required to show workings.)

Write the matrices **P** and **D** in the box (give your answers in fractions)

P = **D =**

(iii) (3 marks) Use diagonalization to find the market shares of the three brands in the long run. Show your workings.

Working space

Write the answer in the box

a_n =

b_n =

c_n =

(iv) (2 marks) True or false: Regardless of the initial market shares a_0, b_0 and c_0, it is not possible for the three coffee brands to have equal market shares in the long term.

Working space

Write either True or False in the box

Additional working space for Q18 (Indicate the parts clearly.)

Question 19 (13 marks)

(a) Given SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} -5 & 4 \\ -4 & 5 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ where the underlying matrix has two linearly independent eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

(i) (3 marks) Write down a fundamental set of solutions of the SDE using the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

Working space

Write the answer in the box

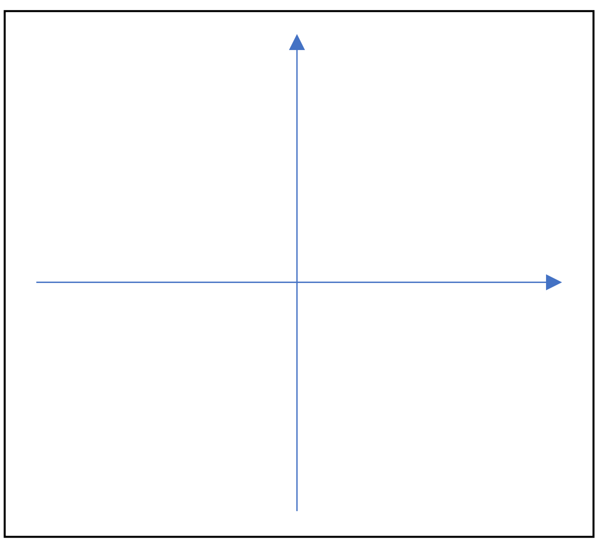
(ii) (3 marks) Solve the initial value problem with the above SDE and the initial values $y_1(0) = 3$ and $y_2(0) = 0$.

Working space

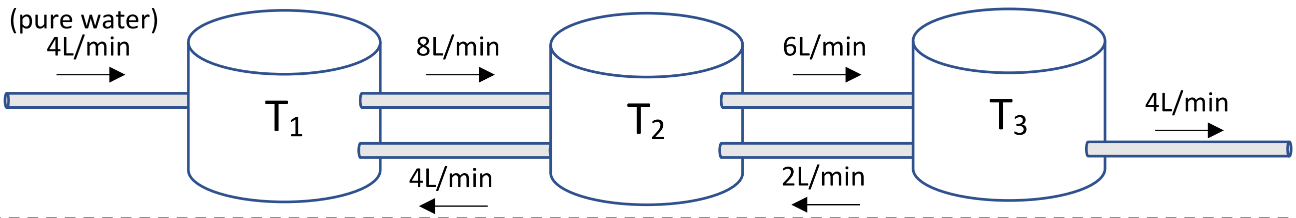
Write the answer in the box

(iii) (3 marks) Sketch the phase portrait of the above SDE.

Sketch the phase portrait in the box



(b) (4 marks) Set up a 3 x 3 SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$ of the following mixing tank problem with variables y_1, y_2 and y_3 representing the amount of salt in the three tanks at time t respectively. The amount of water in each tank is 100 litre (L). (You are <u>not required</u> to solve the SDE.)



<u>Working space</u>

Write the underlying matrix in the box

A =

Additional working space for Q19 (Indicate the parts clearly.)

Additional writing space (Indicate the question and part numbers clearly.)